Front-End Website URL: http://classwork.engr.oregonstate.edu:8742

Feedback Change Log

We have received peer feedback reviews for every draft submission step for this project thus far. Below are summaries of that feedback and the actions we have taken to implement that feedback, for each step of the project:

Step 3 Draft Feedback

Reviewers: Alexa Jauregui, Trisha Rajesh, Adam Fitzpatrick, Kira Stephenson.

Actions Based on Feedback

Based on the peer reviews above and our TA grader, we made the following corrections:

1. **Swap Order of Feedback and Main Content** (TA): Informed that we should switch around the Feedback Change Log and the Main Content sections for this proposal PDF.

Feedback to Implement in Step 4

Based on the peer reviews above and our TA grader, these are the changes we will make on Step 4:

1. **Add a validatorAdminName attribute to RunSubmissions** (Kira Stephenson): Add an additional attribute to the RunSubmissions table for the name of the admin who validated a run.
2. **Meaningful Foreign Key Input Dropdowns** (Adam Fitzpatrick): Update all instances of foreign keys on other tables to possibly reference some key value in the dropdown box form when updating or adding the foreign key, rather than displaying the foreign key integer value. This will improve usability, since the user won't have to cross reference other tables to get the correct key value. However, be sure that the underlying logic continues to use the integer foreign key to connect the entities together.

Step 2 Feedback

Reviewers: Rebecca Charles, Justin Lee, Ciaran Szwarc, Timothy Birmingham.

Actions Based on Feedback

Based on the peer reviews above, we made the following corrections:

1. **Cleanup Schema and ER Diagram** (All Reviewers): Shifted the positioning of some ables on both the Schema and ER diagrams to ensure that all entity connections are clear and easy to read, and the tables aren't overlapping the connection lines.

Additional Design Changes for Step 3

- **Updated Proposal Document**: Updated this document to include any new information, and adjusted some language phrasing here and there in the document to make things more clear. Also decreased information redundancy.
- **Data Manipulation SQL Queries**: Created an SQL queries file that implements the SELECT, INSERT, UPDATE, and DELETE queries. These queries will be used later in future steps of the project.
- **Front-End Website Pages**: Created the base structure for the frontend of the project, using the Astro framework. Created pages for each of the entities in the database that display information from those entity tables.
- **Removed Attributes on ERD**: Chose to remove additional attributes on the ERD that didn't have to do with entity relationships, saving this information for the more detailed schema diagram.

Step 1 Feedback

Reviewers: Kain Sherman, Olivia Choi, Bryant Aragon, Lucas Feldsien.

Actions Based on Feedback

Based on the peer reviews above, we made the following corrections:

1. **Naming Consistency – Singular to Plural** (Kain Sherman): Changed all entity names from singular to plural (Game → Games, Player → Players, RunCategory → RunCategories, Platform → Platforms, RunSubmission → RunSubmissions) to follow the rubric's suggested convention.

2. **Intersection Table Naming Inconsistency** (Kain Sherman): Fixed the inconsistency between the ERD and Relationship Summary table. The table previously mentioned "GamePlatforms" while the ERD showed "GameOnPlatform." We standardized to "GamesOnPlatforms" (plural, matching our other entity naming convention) across both the ERD and all documentation.

3. **Game-RunCategory Relationship Wording Confusion** (Kain Sherman, Lucas Feldsien): Both reviewers identified that the relationship description was contradictory – the Game section implied one RunCategory could be associated with many Games, while the RunCategory section stated it could only belong to one Game. We corrected this by clarifying that the relationship is 1:M from Games to RunCategories: one Game can have multiple RunCategories, but each RunCategory belongs to exactly one Game. The gameID FK in RunCategories enforces this constraint.

4. **Numerical Scope – Annual Run Estimates** (Kain Sherman, Olivia Choi): Kain suggested more concrete numbers to scope the system, and Olivia specifically recommended estimating annual run submissions. We added the calculation: with 25–100 players per competition and 1–30 runs per player per year, the system is expected to handle approximately 750–3,000 run submissions annually.

5. **Entity Highlighting in Overview** (Kain Sherman): Added colored highlighting to entity names in the Overview section so they can be quickly mapped to the Database Outline, improving readability as suggested.

6. **Verification Date Attribute** (Olivia Choi): Added `verifiedDate: date` attribute to RunSubmissions to track when a run was verified by an admin, supporting admin workflows and record-keeping as suggested.

7. **User Roles Clarification** (Olivia Choi): Clarified in the Overview that the website will be "editable by admins, and viewable by players," explicitly stating who will use the site and their primary actions as suggested.

8. **Run Category Edge Cases** (Olivia Choi): Olivia asked whether every run submission must always belong to exactly one category. We confirm this is by design: each RunSubmission has a NOT NULL FK to RunCategories, enforcing that every run belongs to exactly one category. This matches standard speedrunning practice where runs are submitted to specific categories with distinct rulesets (e.g., a "100%" run cannot simultaneously be an "Any%" run).

Additional Design Changes for Step 2

- **Normalization Verified**: Applied the normalization process to all tables *(details in the Normalization section below)*. All tables satisfy 3NF.
- **CASCADE Constraints**: Added ON DELETE CASCADE and ON UPDATE CASCADE to all foreign key constraints, as recommended by the course materials.
- **Schema Diagram**: Created based on DDL implementation, generated using phpMyAdmin Designer.
- **Example Data**: Added 3–5 rows of sample data per table that demonstrate all relationship types in action.

Project Overview, Database Outline, ERD, Schema, Example Data – Updated Version

Project Overview

The following overview is a fictional scenario we are using as a framework for our project.

Northern Oregon Speedrunners Association is a nonprofit organization that runs various video game speedrunning events in local communities in Northern Oregon. They would like to have a website where they can store the various speedrun submissions from their competitions, which typically consist of 25–100 players, depending on the competition. The amount of runs submitted per year will vary drastically, depending on activity in the local speedrunning community, but on average most players will submit 1–30 runs per year. Based on these figures, the system is expected to handle approximately 750–3,000 **run submissions** annually.

Due to the limited needs of this system, they don't need a way to categorize the runs by the event they are a part of, and storing them by date is enough information for them to infer the event they are from. This then allows users to submit runs to be verified that they may complete from home by sending the info about their run to an administrator of the association.

The association only wants **run submissions** to be able to be submitted by admin board members of the association. They would like to be able to see the name of the **players**, the runs they have successfully completed, what **game** they speedran, what **run categories** these runs are in, and what **platform** they were on when they completed the run. This information will serve as a record for all local speedrunners in the area to show off their previous times, and gives **Northern Oregon Speedrunners Association** a record to determine local speedrunning records. This website will be editable by admins, and viewable by players.

Note: *To accommodate for players who may have moved to a different part of the world, the player entity will contain a country.*

Database Outline

Players

Description: A speedrunner participant (via a displayName) that can speedrun various games.

- **playerID**: int, auto_increment, unique, not NULL, PK
- **displayName**: varchar(100), not NULL, unique – The player's username/unique handle
- **country**: varchar(100) – The player's country of residence

Relationship(s):

- 1:M with RunSubmissions with playerID as a FK in RunSubmissions (one Player can have many RunSubmissions)

Games

Description: A video game that is able to be speedrun.

- **gameID**: int, auto_increment, unique, not NULL, PK
- **title**: varchar(255), not NULL – The name of the video game
- **releaseYear**: YEAR – Year the game was released
- **developer**: varchar(255) – The company that developed the game

Relationship(s):

- 1:M with RunSubmissions with gameID as a FK in RunSubmissions (one Game can have many RunSubmissions)
- 1:M with RunCategories with gameID as a FK in RunCategories (one Game can have multiple RunCategories)
- M:N with Platforms via *GamesOnPlatforms* intersection table with both gameID and platformID as FKs (one Game can be on many Platforms; one Platform can have many Games)

RunCategories

Description: A speedrun submission type which different games provide.

- **runCategoryID**: int, auto_increment, unique, not NULL, PK
- **name**: varchar(100), not NULL – The category name (e.g., "Any%", "Glitchless", "100%")
- **ruleset**: text – Description of rules for this category
- **gameID**: int, not NULL, FK – References Games.gameID

Relationship(s):

- 1:M with RunSubmissions with runCategoryID as a FK in RunSubmissions (one RunCategory can have many RunSubmissions)
- M:1 with Games with gameID as a FK in this RunCategories table (each RunCategory belongs to one and only one Game)

Platforms

Description: The hardware or environment a game can be played on.

- **platformID**: int, auto_increment, unique, not NULL, PK
- **name**: varchar(100), not NULL, unique – The platform name (e.g., "PC", "Nintendo Switch", "PlayStation 5")

Relationship(s):

- 1:M with RunSubmissions with platformID as a FK in RunSubmissions (one Platform can have many RunSubmissions)
- M:N with Games via *GamesOnPlatforms* intersection table with both platformID and gameID as FKs (one Platform can have many Games; one Game can be on many Platforms)

RunSubmissions

Description: A specific speedrun record that has all the information about that speedrun attempt.

- **runSubmissionID**: int, auto_increment, unique, not NULL, PK
- **runTime**: time(3), not NULL – The duration of the speedrun (HH:MM:SS.mmm)
- **submissionDate**: date, not NULL – The date the run was submitted
- **verified**: boolean, default FALSE – Whether the run has been verified by an admin
- **verifiedDate**: date – The date the run was verified (NULL if not yet verified)
- **playerID**: int, not NULL, FK – References Players.playerID
- **gameID**: int, not NULL, FK – References Games.gameID
- **platformID**: int, not NULL, FK – References Platforms.platformID

- **runCategoryID**: int, not NULL, FK – References RunCategories.runCategoryID
- **videoLink**: varchar(2048) – URL to video proof of the run

Relationship(s):

- M:1 with Players with playerID as a FK in this RunSubmissions table (many RunSubmissions belong to one Player)
- M:1 with Games with gameID as a FK in this RunSubmissions table (many RunSubmissions belong to one Game)
- M:1 with Platforms with platformID as a FK in this RunSubmissions table (many RunSubmissions were played on one Platform)
- M:1 with RunCategories with runCategoryID as a FK in this RunSubmissions table (many RunSubmissions are in one RunCategory)

GamesOnPlatforms *(Intersection Table)*

Description: Tracks which games are available on which platforms (M:N relationship).

- **gameOnPlatformID**: int, auto_increment, unique, not NULL, PK
- **gameID**: int, not NULL, FK – References Games.gameID
- **platformID**: int, not NULL, FK – References Platforms.platformID

Constraint: Unique combination of (gameID, platformID)

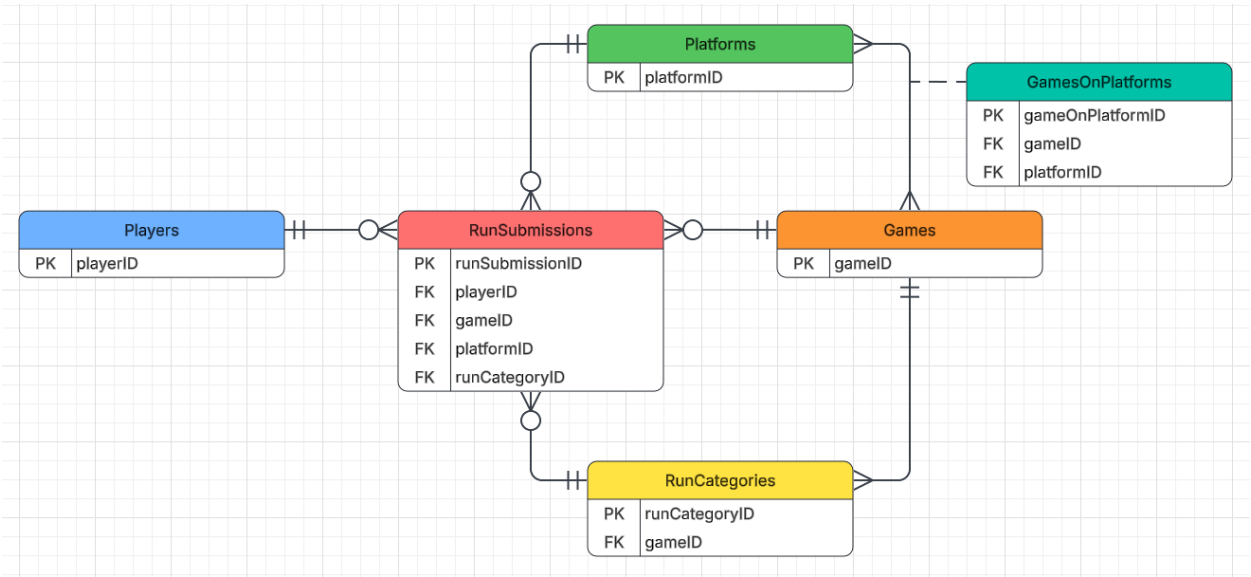
Relationship Summary

Relationship	Type	Implementation
Players → RunSubmissions	**1:M**	playerID FK in RunSubmissions
Games → RunSubmissions	**1:M**	gameID FK in RunSubmissions
Platforms → RunSubmissions	**1:M**	platformID FK in RunSubmissions
RunCategories → RunSubmissions	**1:M**	runCategoryID FK in RunSubmissions
Games ↔ Platforms	**M:N**	GamesOnPlatforms intersection table
Games → RunCategories	**1:M**	gameID FK in RunCategories

Total: 6 relationships (including 1 M:N) across 6 tables

Entity-Relationship Diagram

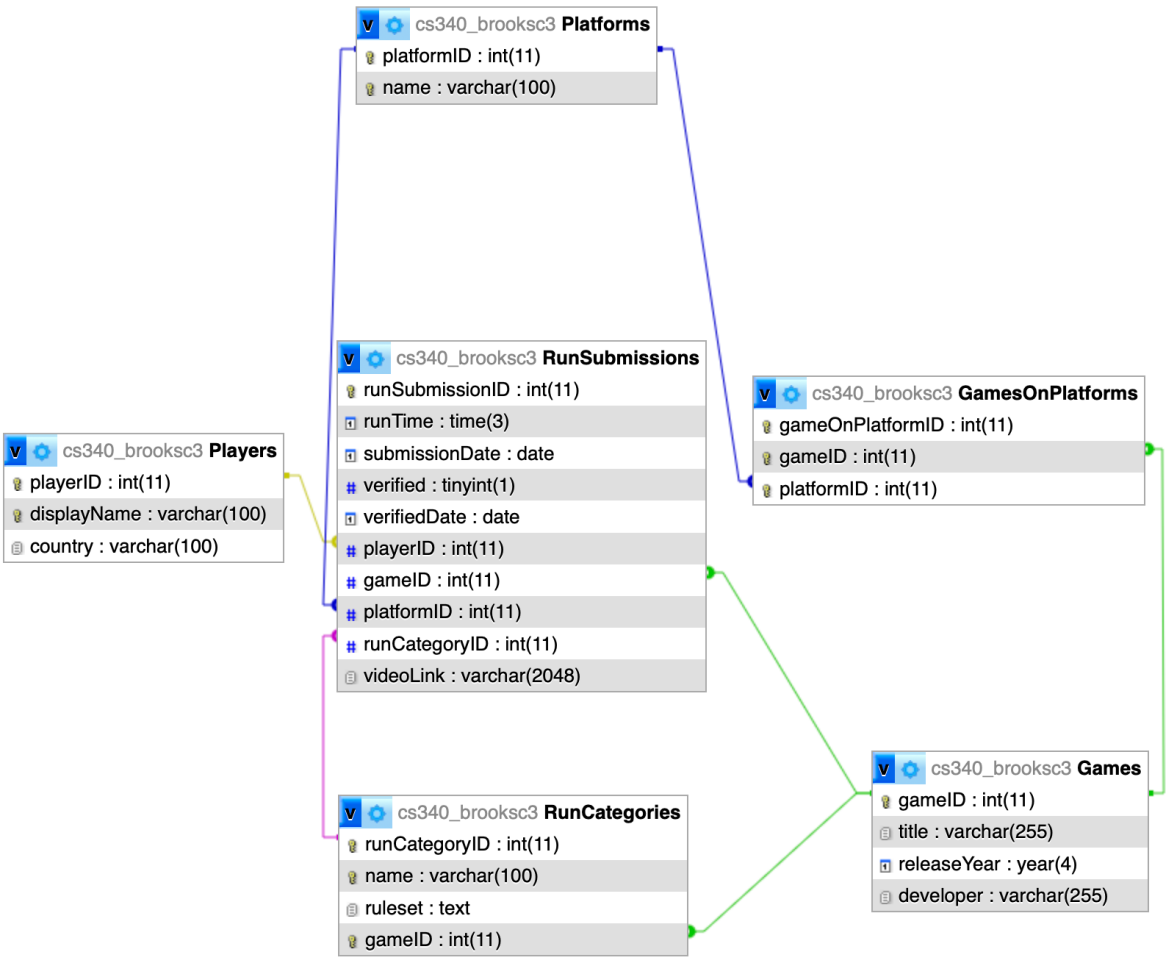
Demonstrates a visual representation of the relationships between the entities listed above. Only primary keys are shown in the ERD; full attributes are listed in the Database Outline above and the Schema Diagram below.



Note: The ERD matches the Database Outline and Schema Diagram. All entity names are plural, the intersection table is named GamesOnPlatforms, and the verifiedDate attribute has been added to RunSubmissions.

Schema Diagram

The schema diagram below was generated from phpMyAdmin Designer after importing the DDL into the CS340 class database on the ENGR server. It shows all tables, their columns, primary keys, and foreign key relationships.



Note: This schema matches the Database Outline and ERD exactly. All foreign keys use ON DELETE CASCADE and ON UPDATE CASCADE constraints.

The following tables provide a detailed textual reference for each table's attributes and constraints:

Players		
Attribute	**Type**	**Constraints**
playerID	int	PK, auto_increment, NOT NULL
displayName	varchar(100)	NOT NULL, UNIQUE
country	varchar(100)	

Games		
Attribute	**Type**	**Constraints**
gameID	int	PK, auto_increment, NOT NULL
title	varchar(255)	NOT NULL
releaseYear	YEAR	
developer	varchar(255)	

Platforms		
Attribute	**Type**	**Constraints**
<u>platformID</u>	int	PK, auto_increment, NOT NULL
name	varchar(100)	NOT NULL, UNIQUE

RunCategories		
Attribute	**Type**	**Constraints**
<u>runCategoryID</u>	int	PK, auto_increment, NOT NULL
name	varchar(100)	NOT NULL
ruleset	text	
gameID	int	FK → Games.gameID, NOT NULL

RunSubmissions		
Attribute	**Type**	**Constraints**
<u>runSubmissionID</u>	int	PK, auto_increment, NOT NULL
runTime	time(3)	NOT NULL
submissionDate	date	NOT NULL
verified	boolean	DEFAULT FALSE
verifiedDate	date	(NULL if unverified)
playerID	int	FK → Players.playerID, NOT NULL
gameID	int	FK → Games.gameID, NOT NULL
platformID	int	FK → Platforms.platformID, NOT NULL
runCategoryID	int	FK → RunCategories.runCategoryID, NOT NULL
videoLink	varchar(2048)	

GamesOnPlatforms *(Intersection Table)*		
Attribute	**Type**	**Constraints**
<u>gameOnPlatformID</u>	int	PK, auto_increment, NOT NULL
gameID	int	FK → Games.gameID, NOT NULL
platformID	int	FK → Platforms.platformID, NOT NULL
UNIQUE constraint on (gameID, platformID)		

Schema Relationships

Players	$\xrightarrow{1:M}$	RunSubmissions
Games	$\xrightarrow{1:M}$	RunSubmissions
Games	$\xrightarrow{1:M}$	RunCategories
Platforms	$\xrightarrow{1:M}$	RunSubmissions
RunCategories	$\xrightarrow{1:M}$	RunSubmissions
Games	M:N	Platforms *(via GamesOnPlatforms)*

Normalization Analysis

We applied the normalization process that we learned about in the Exploration – Normalization Steps. We started by examining a hypothetical sample report (similar to the construction company example from the exploration) and checked for redundancies and anomalies.

Hypothetical Denormalized Report

Consider a report of various speedruns that an administrator would want to add to the database, listing all run submissions based on the following criteria:

Player	Country	Game	Developer	Category	Platform	RunTime	Verified
SpeedDemon42	US	Mario 64	Nintendo	Any%	N64	15:35	Yes
SpeedDemon42	US	Mario 64	Nintendo	16 Star	N64	49:12	Yes
PixelRunner	CA	Zelda: OoT	Nintendo	Any%	Switch	18:22	No
GlitchHunter	US	Celeste	MaddyMakes	Any%	PC	32:45	Yes
NightOwlPlays	JP	Portal	Valve	Inbounds	PC	14:08	No

Redundancy and Anomaly Analysis

If all data were stored in a single flat table:

- **Redundant Data**: "SpeedDemon42" and "US" are repeated for each run by that player. "Nintendo" is repeated for each game developed by Nintendo. "Super Mario 64" is repeated for each run of that game.
- **Insert Anomaly**: To add a new game, we would need to create a dummy run submission. To add a new player, we would need a dummy run.
- **Update Anomaly**: Changing a player's country would require updating every row with that player. Changing a game's developer would require updating every run for that game.
- **Delete Anomaly**: Deleting the only run for a game would lose all information about that game. Deleting the only run for a player would lose that player's information.

1NF Verification

All tables in the final Schema are in 1NF:

- Every table has a defined primary key (no repeating groups).
- All attributes contain atomic values (single value per cell, no lists or sets).
- No table has repeating groups – each row is uniquely identifiable by its PK.

2NF Verification

All tables in the final Schema are in 2NF:

- No table has a composite primary key with partial dependencies. Players, Games, Platforms, RunCategories, and RunSubmissions all use single-column auto-increment primary keys, so partial dependency is impossible.
- The GamesOnPlatforms table contains no non-key attributes beyond the two foreign keys, so there are no partial dependencies.

3NF Verification

All tables in the final Schema are in 3NF:

- **Players**: `playerID` → `displayName`, `country`. No transitive dependency (country does not determine any other non-key attribute).
- **Games**: `gameID` → `title`, `releaseYear`, `developer`. No transitive dependency (developer is an independent non-key attribute and does not functionally determine `title` or `releaseYear`; multiple games can share the same developer while having different titles and release years, so there is no transitive dependency among non-key attributes).
- **Platforms**: `platformID` → `name`. Only one non-key attribute; no transitive dependency possible.
- **RunCategories**: `runCategoryID` → `name`, `ruleset`, `gameID`. The gameID is a FK, not a transitive determinant. The name and ruleset are specific to the category, not determined by gameID alone (one game has multiple categories with different names and rulesets).
- **RunSubmissions**: `runSubmissionID` → all other attributes. All non-key attributes (runTime, submissionDate, verified, verifiedDate, videoLink) are facts about the specific run submission, not about any other non-key attribute. The FKs (playerID, gameID, platformID, runCategoryID) reference other tables – they do not transitively determine each other.
- **GamesOnPlatforms**: Contains only the surrogate PK and two FKs. No non-key attributes to create transitive dependencies.

Conclusion: All six tables satisfy 3NF. No denormalization decisions were necessary in Step 2, as the schema was already properly decomposed from the initial design in Step 1.

Example Data

The following tables show the sample data inserted into each table. This data is included in the accompanying DDL.sql file via INSERT statements. The data demonstrates how foreign keys

enforce the 1:M relationships between the entities, and how the intersection table shows the M:N relationship in action.

Players (5 rows)

playerID	displayName	country
1	SpeedDemon42	United States
2	PixelRunner	Canada
3	GlitchHunter	United States
4	NightOwlPlays	Japan
5	TurboTina	United Kingdom

Games (4 rows)

gameID	title	releaseYear	developer
1	Super Mario 64	1996	Nintendo
2	The Legend of Zelda: Ocarina of Time	1998	Nintendo
3	Celeste	2018	Maddy Makes Games
4	Portal	2007	Valve

RunCategories (5 rows)

runCategoryID	name	ruleset	gameID (FK)
1	Any%	Complete the game as fast as possible by any means.	1
2	16 Star	Complete the game collecting exactly 16 stars.	1
3	Any%	Complete the game as fast as possible by any means.	2
4	Any%	Complete the game as fast as possible by any means.	3
5	Inbounds	Complete all chambers without leaving intended boundaries.	4

Note: Game 1 (Super Mario 64) has two categories (Any% and 16 Star), demonstrating the 1:M relationship from Games to RunCategories. Each category name like "Any%" can appear for different games with different rulesets, but each row belongs to exactly one game.

Platforms (4 rows)

platformID	name
1	PC
2	Nintendo 64
3	Nintendo Switch
4	PlayStation 5

Note: PlayStation 5 (platformID=4) has no games in GamesOnPlatforms, demonstrating that a Platform can exist without associated games.

RunSubmissions (5 rows)

ID	runTime	subDate	verified	verifiedDate	playerID	gameID	platID	catID
1	00:15:35.230	2026-01-10	TRUE	2026-01-11	1	1	2	1
2	00:49:12.800	2026-01-15	TRUE	2026-01-16	1	1	2	2
3	00:18:22.450	2026-01-20	FALSE	NULL	2	2	3	3
4	00:32:45.110	2026-02-01	TRUE	2026-02-02	3	3	1	4
5	00:14:08.670	2026-02-03	FALSE	NULL	4	4	1	5

Note: Column headers abbreviated for readability. Full column names: runSubmissionID, runTime, submissionDate, verified, verifiedDate, playerID, gameID, platformID, runCategoryID. The videoLink column is omitted from the table for space but is included in the DDL.sql file.

GamesOnPlatforms (7 rows) – Intersection Table

gameOnPlatformID	gameID (FK)	platformID (FK)
1	1	2
2	1	3
3	1	1
4	2	2
5	2	3
6	3	1
7	4	1

M:N demonstration: Game 1 (Super Mario 64) appears on 3 platforms (N64, Switch, PC). Platform 1 (PC) hosts 3 games (Super Mario 64, Celeste, Portal). This shows the many-to-many relationship in action.

Relationship demonstrations:

- **1:M Players→RunSubmissions**: Player 1 (SpeedDemon42) has 2 runs (IDs 1 and 2), while Player 5 (TurboTina) has 0 runs.

- **1:M Games→RunSubmissions**: Game 1 (Super Mario 64) has 2 runs (IDs 1 and 2).
- **1:M RunCategories→RunSubmissions**: Each run belongs to exactly one category.
- **NULL verifiedDate**: Runs 3 and 5 are unverified, so verifiedDate is NULL.
- **NULL videoLink**: Run 4 has no video link, showing that videoLink is optional.

Citations

All of the actual content in this proposal is our original work.

ChatGPT was used to help restructure the document and convert the content into LaTeX, so we could quickly update the changes of the document during Step 3 to keep the visual style consistent from previous project steps. The resulting LaTeX was then manually proofread and fixed by hand.

Prompt Used: *"Could you take the following proposal document, and convert the entire contents of the document into the LaTeX format?"*

Claude was used to help setup the core of the front-end. We then went through and manually checked and ensured all code setup followed our guidelines, and worked on the front-end based on this base.

Prompt Used: *"Help me set up a database connection for my CS340 project. I'm using Astro with Node.js and need to connect to a MySQL database on the OSU ENGR servers (classmysql.engr.oregonstate.edu). I want to use the mysql2 package with a .env file for credentials."*